

[illegible]

- Discrete distributions
  - $\rightarrow$  Exponential for lifetime of non-repairing objects
  - $\rightarrow$  time between events in poisson process
  - $\rightarrow$   $P\{X=k\} = \frac{1}{k!} e^{-\lambda} \lambda^k$
  - $\rightarrow$   $e^{-\lambda} = P\{X=0\}$  (no events in time  $\lambda$ )
  - $\rightarrow$   $P\{X \leq k\} = 1 - e^{-\lambda} \sum_{i=0}^{k-1} \frac{\lambda^i}{i!}$
  - Moments:  $E[X] = \frac{1}{\lambda}$      $Var[X] = \frac{1}{\lambda^2}$
  - Poisson distribution:  $O(\lambda) = S^{-1}(e^{-\lambda} \sum_{i=0}^{k-1} \frac{\lambda^i}{i!})$
- $\rightarrow$   $Y = B(X, \lambda)$ 
  - instead proportional to power of time
  - $\rightarrow$   $P\{X=k\} = \frac{1}{k!} e^{-\lambda} \lambda^k$  with  $\lambda = \lambda_1 + \dots + \lambda_n$
  - $\rightarrow$   $X_1, \dots, X_n$  stoch. indep.  $\rightarrow$   $P\{X=k\}$  stoch. indep.
  - $\rightarrow$   $S^{-1}(e^{-\lambda} \sum_{i=0}^{k-1} \frac{\lambda^i}{i!}) = \exp(-\lambda_1) \sum_{i=0}^{k-1} \frac{\lambda_1^i}{i!} \dots$  see 4.4
  - $\rightarrow$   $E[X] = \frac{1}{\lambda} = \frac{1}{\lambda_1 + \dots + \lambda_n}$      $Var[X] = \frac{1}{\lambda^2} = \frac{1}{(\lambda_1 + \dots + \lambda_n)^2}$
- $\rightarrow$  Quantile function:  $O(\lambda) = S^{-1}(e^{-\lambda} \sum_{i=0}^{k-1} \frac{\lambda^i}{i!})$
- $\rightarrow$   $L(X, \lambda) = L(X, \lambda_1) + \dots + L(X, \lambda_n)$
- hump shaped hazard
  - $S(t) = 1 - \frac{t}{\theta} \exp(-\frac{t}{\theta})$
  - $F(t) = 1 - S(t) = \frac{t}{\theta} \exp(-\frac{t}{\theta})$
  - $f(t) = \exp(-\frac{t}{\theta})$

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| -Var( $\theta$ ) = $\text{Exp}(\theta^2) - (\text{Exp}(\theta))^2$                                                                                                                                                                          | $\Rightarrow H(\theta) =$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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| - <b>Censoring</b>                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          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| - <b>Right-censoring</b> - does not observe Y, but C ends<br>- "the observation period"<br>- Y ~ VF event time, C ~ Exp Censoring time<br>- observe T = min(Y, C)<br>- define $\delta = \begin{cases} 1, & T = Y \\ 0, & T = C \end{cases}$ | 4 phases<br>1st phase<br>2nd phase<br>3rd phase<br>4th phase<br>5th phase<br>6th phase<br>7th phase<br>8th phase<br>9th phase<br>10th phase<br>11th phase<br>12th phase<br>13th phase<br>14th phase<br>15th phase<br>16th phase<br>17th phase<br>18th phase<br>19th phase<br>20th phase<br>21st phase<br>22nd phase<br>23rd phase<br>24th phase<br>25th phase<br>26th phase<br>27th phase<br>28th phase<br>29th phase<br>30th phase<br>31st phase<br>32nd phase<br>33rd phase<br>34th phase<br>35th phase<br>36th phase<br>37th phase<br>38th phase<br>39th phase<br>40th phase<br>41st phase<br>42nd phase<br>43rd phase<br>44th phase<br>45th phase<br>46th phase<br>47th phase<br>48th phase<br>49th phase<br>50th phase<br>51st phase<br>52nd phase<br>53rd phase<br>54th phase<br>55th phase<br>56th phase<br>57th phase<br>58th phase<br>59th phase<br>60th phase<br>61st phase<br>62nd phase<br>63rd phase<br>64th phase<br>65th phase<br>66th phase<br>67th phase<br>68th phase<br>69th phase<br>70th phase<br>71st phase<br>72nd phase<br>73rd phase<br>74th phase<br>75th phase<br>76th phase<br>77th phase<br>78th phase<br>79th phase<br>80th phase<br>81st phase<br>82nd phase<br>83rd phase<br>84th phase<br>85th phase<br>86th phase<br>87th phase<br>88th phase<br>89th phase<br>90th phase<br>91st phase<br>92nd phase<br>93rd phase<br>94th phase<br>95th phase<br>96th phase<br>97th phase<br>98th phase<br>99th phase<br>100th phase<br>101st phase<br>102nd phase<br>103rd phase<br>104th phase<br>105th phase<br>106th phase<br>107th phase<br>108th phase<br>109th phase<br>110th phase<br>111st phase<br>112nd phase<br>113th phase<br>114th phase<br>115th phase<br>116th phase<br>117th phase<br>118th phase<br>119th phase<br>120th phase<br>121st phase<br>122nd phase<br>123rd 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- Interval Censoring
  - event happened in time interval, but exact time is unknown
- Left Censoring
  - Event happened earlier than first time point but we don't know how much earlier
- Right Censoring
  - individuals not included in study because event happened earlier
  - e.g. infant mortality, but only babies that are already one year old followed by multi-state transition
  - right censoring
    - event has occurred but condition that it is in the data is not fulfilled
    - e.g. Posing reporting times
- > Censoring: When sent the Person, also write willingness to turn over

Transition: When the Person leaves all states they enter "non-response" (left set) -> if you don't observe them before or after their death status = Data missing

=> E(t)

=> U(t-)

=> V(t-)

=> W(t-)

W(t-) < U(t-) < V(t-)

U(t-) < V(t-)

tapering out

33) Plot likelihood

- Y(t)=y\_i

- P(y\_i=0)

- N(y\_i=0)

=> L(t)

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$t_{\text{Surv}} = \frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}} = 2 \cdot 400$   
 phase (sampled Survival days, studies)  $\sim 1$   
 data (sampled Survival days for computational  
 F-Estimator Row  $S_i = S_{ij}(t) = \exp(-C(t))$   
 - non-parametric  
 - time distributions  
 - Test for 2 groups  
 - null-hypothesis  $H_0: C_1 = C_2$   
 - number of events in group at  $t_j$   
 - number of individuals at risk in group at  $t_j$   
 $\frac{d_{\text{Surv}}}{n_{\text{Surv}}}$  individuals randomly drawn from  $n_{\text{Surv}}$

a)  $P(t_j) = \frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 b) Lognormal Model  
 $\sim \frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 $\sim P(t_j) \cdot \frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 $S_{\text{Surv}}(t_j) = P(t_j) \cdot F(t_j) = C(t_j)$   
 c)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 d)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 e)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 f)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 g)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 h)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 i)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 j)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 k)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 l)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 m)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 n)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 o)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 p)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 q)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 r)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 s)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 t)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 u)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 v)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 w)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 x)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 y)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$   
 z)  $\frac{d_{\text{Surv}}}{n_{\text{Surv}}} \cdot n_{\text{Surv}}$

$\hat{\theta}_n = \theta_0 + \frac{1}{n} \sum_{i=1}^n \epsilon_i$   
 $E(\hat{\theta}_n) = \theta_0$   $Var(\hat{\theta}_n) = \frac{1}{n} Var(\epsilon_1)$   
 sample covariance matrix with  $n$  rep. expts  
 $(\hat{\Sigma}_n, \hat{\mu}_n)$  not absolute because of test  
 but  $\hat{\Sigma}_n$  is not absolute because of test  
 because not same: known steps  
 $\hat{\Sigma}_n = \frac{1}{n} \sum_{i=1}^n (\epsilon_i - \bar{\epsilon})(\epsilon_i - \bar{\epsilon})^T$   
 $\hat{\mu}_n = \bar{\epsilon}$   
 test  
 not suitable by use of risk test  
 $(d_n, \epsilon_n)$  test  
 $\frac{1}{n} \sum_{i=1}^n \epsilon_i$   
 $\frac{1}{n} \sum_{i=1}^n \epsilon_i$   
 level on information at beginning  
 coarse estimation  $\rightarrow$  **coarsening**  
 estimation and coarsening  
 $\hat{\theta}_n$  with realizations  $\theta_1, \dots, \theta_n$   
 usually **sampling distribution**  
 usually **joint**  $\rightarrow$  **inferential distribution**  
 parameter vector  
 off  $\hat{\theta}_n$  in  $\theta_0$

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 $\hat{\Sigma}_n = \frac{1}{n} \sum_{i=1}^n (\epsilon_i - \bar{\epsilon})(\epsilon_i - \bar{\epsilon})^T$   
 $\hat{\mu}_n = \bar{\epsilon}$   
 test  
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- Intercepts**
  - $\exp(\beta_0) = \exp(\beta_1)$
  - identical for all units
  - there is exact at  $\frac{\beta_1}{\beta_0} = \frac{\exp(\beta_1)}{\exp(\beta_0)}$
  - $\frac{\beta_1}{\beta_0} = \frac{\exp(\beta_1)}{\exp(\beta_0)}$
  - proportion that only want per time point
  - $\frac{\exp(\beta_1)}{\exp(\beta_0)} = 1$
- Denominate Residuals**
  - $y_{it} - \text{sign}(\beta_1) \sqrt{2} \cdot [e^{\beta_1}] \cdot \exp(\beta_0)$
  - for detection of outliers and to see whether to include numeric linearly or not
- Standardized residuals**
  - used to get the specific variable
  - $\frac{y_{it} - \text{sign}(\beta_1) \sqrt{2} \cdot [e^{\beta_1}] \cdot \exp(\beta_0)}{\exp(\beta_0)}$
  - with PH Assumption:  $E[\epsilon_{it}] = 0$
  - plot  $\frac{y_{it} - \text{sign}(\beta_1) \sqrt{2} \cdot [e^{\beta_1}] \cdot \exp(\beta_0)}{\exp(\beta_0)}$  vs  $t$  should have no trend
  - for checking PH Assumption (const.)
  - $\text{cor}(\beta_1, \text{ph model}, \text{transform})$
  - $\beta_1$  should be constant over time
  - $p\text{-value} > 0.05$
- Cox-Norm-Ph** (Covariate change)
  - Time-dependent Covariates

[illegible]

|                                                     |                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\log \text{PL}(\beta_1) - \log \text{PL}(\beta_0)$ | log-likelihood ratio test for sufficient levels of stratification. variable $x$ :<br>$h(x; \beta_1, \beta_0) = h_0(x; \beta_0) \cdot \exp(x' \beta)$                                                                                                                                                            |
| of enhance                                          | $h_0(x; \beta_0)$ : risk set of events of $x$ at $t_0$<br>$\rightarrow h_0(x; \beta_0) < \infty \Rightarrow x_k \in \mathcal{X}_k$ and $x_j = 0_j$<br>$\mathcal{X}_k = \{x_1, \dots, x_n\}$<br>$\Rightarrow \text{PL}(\beta) = \frac{1}{n} \sum_{k=1}^K \frac{\exp(x_k' \beta)}{\sum_{j=1}^n \exp(x_j' \beta)}$ |
| by adjusting Risk-set                               | $\rightarrow \text{Breslow-Holm}(\beta; t_0) = \frac{1}{\sum_{j=1}^n \exp(x_j' \beta)}$                                                                                                                                                                                                                         |
| $t_0 = \tau_0(x_0)$                                 | Piecewise Exponential Model (PEM)<br>- transforms survival data into standard regression data transformation & Post processing                                                                                                                                                                                  |

|                                            |                                                                                                                                                                                                                                                                                                                                                                 |                                    |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| $B(\lambda)$                               | <b>Pre-define Exponential Data:</b><br>$\lambda$ as $\text{prob}(\text{failure}   \text{hour})$                                                                                                                                                                                                                                                                 | <b>Cut-off times:</b><br>remembers |
| exp. lifetimes                             | - internal time line for each subject as it<br>- internal time the subject is at risk<br>- subjects not necessarily enter times<br>- lowest index $i$ to $j$ else<br>- time spent $t_j - t_{j-1}$<br>- end of interval: $t_j = t_{j-1} + \Delta t$                                                                                                              |                                    |
| $\frac{1}{n} \sum_{i=1}^n \mathbb{E}(t_i)$ | - Post-intervals: stable but rough<br>- many intervals, but not too many<br>- smaller intervals when hazard<br>- choose number of intervals as $p$                                                                                                                                                                                                              |                                    |
| $\mathbb{E}(t_i)$                          | - assume <b>pre-define constant</b> hazard<br>- $\lambda(t) = \lambda_0 \forall t \in [t_{j-1}, t_j]$<br>$\Rightarrow h(t) = \lambda_0 \exp(-\lambda_0 t)$<br>$\Rightarrow \text{hazard} = L(P, \beta) = \tilde{A}(t, \beta) \exp(\beta_0 + \beta_1 t)$<br>$\Rightarrow h(t) = h_0^* \exp(\beta_1 t)$<br>$\Rightarrow \lambda(t) = \lambda_0^* \exp(\beta_1 t)$ |                                    |
| $\frac{1}{n} \sum_{i=1}^n \mathbb{E}(t_i)$ |                                                                                                                                                                                                                                                                                                                                                                 |                                    |

[illegible]

- baseline:  $\mu$
- Penalize Additive Mixed Model (PAMM)
  - $h(\mathbf{x}) = \mathbf{x}^T \beta + \exp(\beta_0) \cdot h_1(\mathbf{x}_1) \cdot \dots \cdot h_p(\mathbf{x}_p)$
  - $\beta$  is the overall intercept of step function in PAMM
  - baseline:  $\mu = \exp(\beta_0)$  →  $\mu$  is the overall mean baseline
  - $h_i(\mathbf{x}_i) \sim \text{family}(\text{Poisson}, \text{link} = \text{offset} + \text{effect})$
  - $\text{status} \sim \text{Poisson}(\lambda = h(\mathbf{x}))$
- measuring effects are just interactions between time and other covariates
- $\text{model}(\text{status} \sim \text{time} + \text{other covariates})$
- $\text{get}(\text{coef}(\text{model}), \text{family}(\text{Poisson}, \text{link} = \text{offset} + \text{effect}))$
- if I just know, estimate from data:  $\text{pammm}(\text{status} \sim \text{time} \sim \text{SE}, \text{type} = \text{c})$
- look at  $\text{coef}$  to check for non-linearity
- if interaction, non-linear and  $\beta$  unknown:  $\text{model}(\text{status} \sim \text{time} + \text{effect}, \text{c})$
- $\text{Shrink}(\text{fit})$ 
  - $\text{shrink}(\text{model}(\text{status} \sim \text{interaction}(\text{time}, \text{c})), \text{c})$
- $\text{Wofe}(\text{coef}, \text{shape} = \text{separate baseline function})$
- estimated  $\rightarrow \text{fit}(\text{type} = \text{c})$
- $\text{log} - \text{transformation}$
- $\text{eff} - \text{inflect}(\text{model}, \text{time} = \text{fit}, \text{ns} = 6)$
- $\text{as}(\text{plot}(\text{Surv}(\text{time}, \text{event})), \text{ns})$
- check whether left truncation time is

independent by including it in the model and it is not significant

random effect:  $b_{ij} = \exp(\alpha_0 + \dots + \alpha_5)$

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Modelling discrete time-to-event data

- discretisation of continuous time scale
- Independent continuous time  $T$  in  $D_{\text{obs}}$   $\leftarrow \alpha_0$
- Discrete time  $T$ :  $T_1 < T_2 < \dots < T_n$
- discrete time hazard

$$h_{ij}(t_k) = P(T_{ij} = t_k)$$

discrete time survival curve

$$S_{ij}(t_k) = P(T_{ij} > t_k) = \prod_{l=1}^k (1 - h_{ij}(t_l))$$

• of exponentiating event at  $t_k$

[illegible]

